

VORONEZH, Russian Federation

WMO: 341230

Lat: 51.700N

Lon: 39.217E

Elev: 149

StdP: 99.55

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-22.9	-19.7	-26.0	0.4	-22.5	-23.1	0.5	-19.2	9.2	-2.7	8.2	-3.9	1.8	30	0.505

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.9	32.2	20.0	30.2	19.4	28.4	18.6	21.5	28.9	20.6	27.5	19.8	26.2	2.9	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.2	14.2	23.7	18.4	13.5	23.1	17.6	12.8	22.7	63.1	29.0	60.1	27.6	57.4	26.2	28.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.9	6.9	6.1	DB	-25.0	34.7	3.6	2.0	-27.6	36.2	-29.7	37.4	-31.7	38.5	-34.3	40.0	(4)
(5)			WB	-25.2	23.4	3.5	1.6	-27.8	24.5	-29.8	25.4	-31.8	26.3	-34.4	27.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	7.6	-6.7	-5.7	-0.3	8.9	15.7	19.2	21.4	20.1	14.1	7.4	0.8	-4.3	(6)	
	DBStd	11.24	6.57	6.44	4.73	4.71	4.64	3.83	3.39	3.87	4.25	4.57	5.20	6.18	(7)	
	HDD10.0	2198	518	439	319	76	8	0	0	0	11	107	277	443	(8)	
	HDD18.3	4191	776	672	577	283	108	36	9	25	137	339	526	702	(9)	
	CDD10.0	1332	0	0	1	43	184	276	354	312	135	26	1	0	(10)	
	CDD18.3	283	0	0	0	1	25	61	106	79	11	0	0	0	(11)	
	CDH23.3	2540	0	0	0	7	245	508	921	755	103	1	0	0	(12)	
	CDH26.7	840	0	0	0	1	58	144	333	286	18	0	0	0	(13)	
(14) Wind	WSAvg	2.7	3.1	3.3	3.2	2.9	2.6	2.4	2.1	2.2	2.3	2.6	3.0	3.2	(14)	
(15) Precipitation	PrecAvg	576	43	34	34	38	45	64	67	58	53	44	48	50	(15)	
	PrecMax	831	96	73	85	78	141	217	197	186	183	139	110	118	(16)	
	PrecMin	379	7	1	4	3	8	6	2	3	5	0	3	12	(17)	
	PrecStd	93	22	17	19	21	29	39	35	32	39	25	30	29	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.0	5.2	14.4	24.4	30.7	32.4	35.3	35.7	29.0	21.3	13.4	8.1	(19)	
		MCWB	3.8	3.7	7.6	13.1	18.7	18.8	20.8	19.9	16.6	14.1	11.0	6.9	(20)	
	2%	DB	3.0	3.7	10.5	21.5	28.3	30.1	32.3	32.3	26.2	18.4	11.1	5.8	(21)	
		MCWB	2.0	2.5	5.8	12.4	16.8	19.1	20.7	19.9	15.9	13.1	9.5	5.0	(22)	
	5%	DB	2.0	2.6	7.8	19.1	26.2	28.2	30.2	29.8	23.8	16.0	8.9	4.2	(23)	
		MCWB	1.1	1.4	4.5	11.3	16.3	18.9	20.1	19.1	15.6	12.2	7.4	3.2	(24)	
	10%	DB	1.1	1.6	5.7	16.8	23.9	26.3	28.2	27.4	21.3	13.8	7.1	2.5	(25)	
		MCWB	0.4	0.7	3.2	10.1	15.6	18.2	19.4	18.2	14.5	11.0	5.9	1.7	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.1	4.2	8.8	14.5	19.8	22.0	23.1	22.1	18.6	15.8	11.5	7.2	(27)	
		MCDB	4.7	4.8	13.4	21.3	28.1	28.1	30.9	30.5	23.5	19.2	13.0	8.1	(28)	
	2%	WB	2.2	2.8	6.3	13.1	18.4	20.7	21.9	20.9	17.2	14.1	9.6	5.1	(29)	
		MCDB	2.8	3.5	9.5	18.8	25.7	26.9	29.6	29.0	23.3	17.2	11.1	5.8	(30)	
	5%	WB	1.2	1.6	4.7	12.0	17.2	19.8	21.0	20.0	16.3	12.6	7.7	3.3	(31)	
		MCDB	1.8	2.3	7.3	17.7	23.7	25.8	27.8	27.4	21.7	15.3	8.8	4.0	(32)	
	10%	WB	0.4	0.8	3.5	10.8	16.2	18.9	20.2	19.1	15.3	11.2	6.1	1.8	(33)	
		MCDB	1.1	1.5	5.5	15.9	22.4	24.6	26.2	25.7	20.0	13.6	7.0	2.4	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	4.9	5.5	6.3	9.2	10.3	9.6	9.9	10.7	9.3	6.6	4.5	4.5	(35)	
		MCDBR	3.5	4.2	9.3	12.1	13.1	12.2	12.9	14.4	13.2	9.7	6.1	3.9	(36)	
	5% WB	MCWBR	3.4	3.8	5.8	6.0	4.9	4.6	4.5	5.0	5.5	5.0	5.0	3.6	(37)	
		MCWBR	3.5	3.8	8.2	10.8	11.5	10.4	11.1	11.7	10.8	8.5	5.6	3.9	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.305	0.341	0.365	0.442	0.398	0.390	0.421	0.436	0.388	0.362	0.338	0.307	(40)	
		taud	2.124	2.026	2.099	2.088	2.315	2.353	2.274	2.216	2.332	2.353	2.276	2.176	(41)	
	Ebn at Noon	Ebn at Noon	667	743	813	791	856	865	828	790	790	734	635	599	(42)	
		Edh at Noon	76	111	126	145	121	118	125	126	101	81	67	62	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	0.96	1.99	3.36	4.17	5.72	5.96	5.84	5.06	3.35	1.77	0.78	0.55	(44)	
	RadStd	RadStd	0.22	0.27	0.54	0.37	0.47	0.52	0.48	0.43	0.45	0.25	0.16	0.15	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	+0.83	N/A	N/A	N/A	N/A	N/A	N/A	-182	+107	N/A	(46)
(47)	Regional (6 neighbors)	+0.74	N/A	N/A	N/A	N/A	N/A	N/A	-184	+116	N/A	(47)

Nomenclature: See separate page